

Frozen sites and active-region statistics in a dipole-conserving lattice chain

Research-progress record

Project: `dipole-sep-hydrodynamics`

Compiled from the repository state at commit 03b6349 plus subsequent uncommitted work

14 September 2026

Abstract

This document reconstructs the current state of the project from the repository: source code, configuration files, raw data, notebooks, the research log, the hypothesis file and the literature notes. Numbers quoted here were recomputed from the raw data rather than copied from the log, and all figures are regenerated from raw data by `numerics/scripts/make_progress_figures.py`. Three verification measurements that the project had listed as open were carried out while preparing this document and are marked [\[new, this document\]](#) throughout; they are also recorded in `RESEARCH_LOG.md`. Interpretations are labelled explicitly and separated from measurements, and Section 6 is a deliberately adversarial assessment of what is weak, unsupported or internally inconsistent in the work so far.

Executive summary

- **Model.** A periodic chain of L sites, $s_i = \pm 1$ at density exactly $1/2$. The only allowed move flips a 4-site window between $(+, -, -, +)$ and $(-, +, +, -)$ (i.e. $1001 \leftrightarrow 0110$), conserving both magnetisation and dipole moment. One uniformly chosen active window flips per step.
- **H1 (frozen fraction) — saturation is well supported.** Across $N = 100$ to 3×10^5 the frozen fraction is flat. A constant fits; logarithmic decay is excluded by a huge margin ($\chi^2_\nu \approx 2.8 \times 10^3$); a pure power law fits only with an exponent consistent with zero, $\alpha = -0.00083 \pm 0.00050$, 95% CI $[-0.0018, +0.00015]$. Best estimates: $f = 0.35988 \pm 0.00015$ (definition A, 31 points) and $f = 0.36303 \pm 0.00065$ (definition B at $W = 5L$, 24 points).
- **The W -dependence worry is now bounded.** [\[new, this document\]](#) Definition B makes the window W part of the observable. Measuring $f_B(W)$ along single trajectories shows it *saturates* for $W \gtrsim 20L$: the production choice $W = 5L$ overshoots the plateau by only $+0.003$ to $+0.004$, and $W = 10L$ by $\lesssim 0.001$.
- **The estimator is validated against ground truth.** [\[new, this document\]](#) Exact breadth-first enumeration of the Krylov sector at $L \leq 60$ gives the *static* frozen fraction that H1 actually defines. Once fully-jammed configurations — which the dynamical estimator structurally cannot sample, and which vanish exponentially with L — are excluded from both sides, exact and dynamical agree with $\chi^2_\nu = 0.83$ over six matched L . This closes the [\[UNCERTAIN\]](#) flag H1 has carried since it was stated.
- **Strong fragmentation, measured directly.** [\[new, this document\]](#) Median Krylov sector size grows as $(1.135)^L$ against $(1.969)^L$ half-filled configurations, so a sector occupies a fraction $\sim (0.576)^L$ of its symmetry sector.
- **H2 (active-region lengths) — exponential, not power law.** The distribution of active-region lengths is exponential with $\lambda \approx 13$ sites, *independent of L* over $L = 100$ – 12800 . A full-range exponential fit beats a power law at all eight L ($R^2 = 0.910$ – 0.966 versus 0.872 – 0.906), and the rescaled distributions collapse. Mean nonzero length 15.4 – 16.1 and zero-gap rate 0.885 – 0.889 are both flat in L .

- **The forbidden lengths are now explained.** [\[new, this document\]](#) Active-region lengths 1, 2, 3 and 5 never occur; 4 and all $\ell \geq 6$ do. An exhaustive finite check (zero violations over all 2^5 configurations) plus a short argument shows this is exact, not statistical: two window positions one site apart can never both fire in isolation. *This derivation has not yet been reviewed by the researcher.*
- **Physical reading (interpretation, not measurement).** A finite frozen fraction with exponentially distributed, L -independent active regions means active clusters stay microscopic as $L \rightarrow \infty$. In the language of the project README, this is the “small clusters” branch: there is no scale-free active structure available to carry an anomalous hydrodynamic tail.
- **Main open weaknesses.** No production run above $N = 10^4$ under the current definition; HYPOTHESES.md still states a static order parameter while every production number is a dynamical proxy; the W -scan settings comparison changed four parameters at once; and the literature’s exact analytic predictions have not been evaluated numerically to compare against our 0.36.

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1 Research question and motivation

1.1 The model

The system is a ring of L sites carrying $s_i \in \{+1, -1\}$ (equivalently occupation $n_i \in \{1, 0\}$), at density exactly $1/2$. Write a 4-site window starting at i as $(s_i, s_{i+1}, s_{i+2}, s_{i+3})$ with indices mod L . The window is *active* iff it equals

$$A = (+1, -1, -1, +1) \quad \text{or} \quad B = (-1, +1, +1, -1), \quad (1)$$

i.e. 1001 or 0110 in occupation language, and the only allowed move is $A \leftrightarrow B$ on an active window. The dynamics is a classical continuous-time-like stochastic process implemented as: at each step, pick one active window uniformly at random and flip it (`numerics/src/dipole_chain.py`).

The move conserves the particle number $Q = \sum_i n_i$ and the dipole moment $P = \sum_i i n_i$ (modulo L under periodic boundaries), because it moves two particles in opposite directions by equal amounts. This is the defining structure of a dipole-conserving / “fracton” lattice gas: single particles are immobile, only correlated pairs move.

A useful exact fact, used below: for a uniformly random half-filled configuration the probability that a given window is active is $2/2^4 = 1/8$, so $\langle n_{\text{active}} \rangle = L/8 = 12.5\%$ of L at $t = 0$; the research log records that this stays at 12.5–13.6% in steady state across $L = 10^2$ – 10^5 .

1.2 Why the problem is interesting

The project README states the motivation directly. Because the kinetic constraint is strong, “some regions do not evolve at all”. The programme is then:

1. establish that a *finite fraction* of the chain is frozen in the thermodynamic limit (not a vanishing one);
2. determine whether the length distribution of the *active* regions between frozen sites is exponential, $\sim e^{-\ell/\eta}$, or a power law, $\sim \ell^{-\alpha}$.

The second question is the one with physical consequences. If the distribution is exponential, active regions are microscopic clusters whose size does not grow with L . If it is a power law, arbitrarily large active clusters survive in the thermodynamic limit, and the README’s target observable — the global density–density correlation function — could acquire an anomalous tail: even with a finite frozen fraction arresting global transport, a perturbation could still relax globally through rare large active regions. The README explicitly contemplates abandoning this model for another if the distribution turns out exponential.

1.3 Hypotheses as stated in the project

H1 (HYPOTHESES.md, 12 Sep)

$f_N \rightarrow f_\infty > 0$ as $N \rightarrow \infty$ (candidate H1a), against H1b $f_N \sim cN^{-\alpha}$ and H1c $f_N \sim c/\ln N$.

The stated order parameter is *static*: a site is frozen iff no sequence of allowed moves starting from the initial configuration can ever change it. The file carries a flagged **[UNCERTAIN]**: this static/dynamical equivalence had never been verified by enumeration.

H2 (14 Sep)

The active-region length distribution either collapses onto an L -independent memoryless form (H2a) or retains an anomalously heavy tail (H2b). The measured quantity is the gap between consecutive frozen sites at a single post-burn-in snapshot, with zero-length gaps discarded by explicit researcher instruction.

Q3 (QUESTIONS.md #3, 14 Sep)

Why do active-region lengths 1, 2, 3, 5 never occur while 4 and all $\ell \geq 6$ do?

1.4 Literature context contained in the project

Seven papers were read and assessed (LITERATURE.md; applicable details in numerics/theory/Notes/). The directly relevant content:

- **Classen-Howes (2024) thesis, Ch. 5 §5.1** treats exactly this model ($d = 2$ hardcore, range $k = 4$). Its general result is a strong-to-weak fragmentation transition at critical filling $\nu_c = 1/(k - 2)$; for $k = 4$ this is $\nu_c = 1/2$, and $(d, k) = (2, 4)$ is one of only two degenerate cases where ν_c coincides with its particle–hole mirror, so the weakly fragmented phase disappears and the model is *strongly fragmented at every filling*. It also derives an exact active-bubble density $\propto \nu^x(1 - \nu)^{\ell-x}$, exponential in ℓ , and gives $\ell_{\min} = \ell_{\max} = 4$ for the minimal two-particle bubble.
- **Morningstar, Khemani & Huse (2020)** study the $n_i \in \{0, 1, 2\}$ version; our model is exactly its hardcore invariant sub-sector, and it uses the same static Krylov-based definition of a frozen site.
- **Moudgalya & Motrunich (2021)** supplies a reusable exact counting technique (canonical form $\rightarrow$ tiling $\rightarrow$ linear recursion).
- Three further papers (Han–Lake–Ro; Glorioso *et al.*; Zerba *et al.*) treat the complementary dense/ergodic regime and do not bear on H1/H2. Sala *et al.* (2020), flagged by three of them as the likely key reference, was checked and found *not* to apply: it is a spin-1 model with a vacancy, and blocks of two like charges — the interior of our only active pattern — are frozen there.

2 Work completed

2.1 Chronology

Date	Stage	Content
12 Sep	H1 v1	Pipeline built; definition A (cumulative) runs to $N = 102400$; fits (constant vs. power-law correction).
13 Sep	H1 v1	fitting_range run to $N = 3 \times 10^5$ (11h29m); combined 31-point fits; H1b/H1c added; overshoot/commensuration checks.
13 Sep	redefinition	“Frozen” redefined as a rolling window (definition B), $W = 5L$, $M = 10$, on researcher instruction.
13 Sep	validation	Fixed-patience burn-in found to fail at large N (2/3 trials drifting at $N = 10^5$); two fixes compared; self-scaling patience (“method 1”) locked in.
13 Sep	H1 v2	production_intermediate $N = 100$ – 10^4 (1h28m); re-run with doubled $W, M, c, n_{\min}$ (6h03m); settings comparison.
14 Sep	H2	Gap-length pipeline built; test (11s) and production ($L = 100$ – 12800 , 14m22s) runs; analysis notebook.
14 Sep	literature	Seven papers reconstructed; LITERATURE.md and Notes/ written.
14 Sep	[new, this document]	$f_B(W)$ scan; exact BFS static frozen fraction; small- L like-for-like validation; forbidden-length argument.

2.2 Three inequivalent definitions of “frozen”

Much of the project’s subtlety lives here, and the three definitions are easy to conflate.

	Definition	Status in the project
Static	Site takes the same value in <i>every</i> configuration reachable by any move sequence. A property of the initial string only.	What HYPOTHESES.md H1 literally states. Never measured until this document; now measured exactly at $L \leq 60$ by BFS.
A (cumulative)	Site has never flipped since $t = 0$; run stopped once no <i>new</i> site has flipped for $20L$ steps.	All 12–13 Sep production data ($N \leq 3 \times 10^5$). Superseded, archived (not deleted).
B (rolling)	Site has not flipped during the trailing window $[t - W, t]$; sites may re-enter the frozen set.	Current production definition; all H1 v2 and all H2 data.

Two structural facts follow immediately and are worth stating because they constrain everything:

$$f_{\text{static}} \leq f_B(W') \leq f_B(W) \quad \text{for } W \leq W', \quad (2)$$

pointwise on any trajectory, since a longer window gives a site more chances to disqualify itself; and definition A is definition B evaluated over the window $[0, T_A]$, so it is directly comparable once T_A is known (measured below: $T_A \approx 27\text{--}32 L$).

W is part of the observable, not the stopping rule. Every H1 v2 number is f_B at one particular W . This was flagged when definition B was adopted but never quantified until Section 3.3.

2.3 Code and computational method

- `src/dipole_chain.py` — the chain, with an $O(1)$ active-site list (swap-with-last) and a local rescan of the ten windows that a flip can affect.
- `src/frozen_fraction.py` — definition A (`measure_frozen_fraction`), definition B (`measure_frozen_fraction_rolling`), the shared self-scaling burn-in `_self_scaling_burnin`, adaptive ensemble drivers.
- `src/burnin_comparison.py` — kept as the record of the method-1 vs. method-2 comparison.
- `src/gap_length.py` — H2 measurement; cyclic gap extraction from the site-level frozen mask.
- [\[new, this document\]](#) `src/w_dependence.py`, `src/static_frozen.py` — the two new measurements, with matching scripts and configs.
- Ensembles use `numpy.random.SeedSequence` spawning with a per-estimator integer tag (1 rolling, 2 gap-length, 3 static, 4 W -scan), so every estimator has an independent, reproducible stream.
- Sampling is adaptive: draw batches until the relative standard error on the mean falls below a target (1% in production) subject to a `min_samples` floor and a cap.

2.4 The burn-in problem, and why it matters

This was the most consequential methodological episode in the project. The original burn-in used a fixed patience of $20L$ steps. The measured growth-phase duration in units of L grows with N , so the ratio of patience to growth time fell from $\sim 5\times$ at $N = 10^2$ to $\sim 0.4\times$ at $N = 10^6$. A trend test across the M measurement windows then found significant drift in 1/5 trials at $N = 3 \times 10^4$ and 2/3 at $N = 10^5$ — i.e. measurements were starting before the system had settled, exactly where the ratio argument predicted.

The adopted fix (“method 1”) extends burn-in in W -length windows until $(t - G) \geq cG$, where G is the step of the most recent first-time flip; this pins the patience/growth ratio at c for every N . It gave 0/10 significant trends at the same N , and agreed with an independent

drift-gated scheme to within ± 0.0004 on identical trajectories. A zero-variance NaN bug in the trend test (which had been silently counted as “not significant”) was fixed in the same pass.

2.5 What is completed, ongoing, and untested

Completed

H1 under definition A up to $N = 3 \times 10^5$ with full model comparison; H1 under definition B up to $N = 10^4$ at two settings; H2 production $L = 100$ –12800 with shape/tail analysis; burn-in validation; estimator validation against exact enumeration; $f_B(W)$ scan; literature review.

Ongoing / configured but not run

`production_large_scale.yaml` ($N = 10^4$ – 2×10^4 , estimated 4.2–4.6 h) is configured and was approved, but a scheduled launch was interrupted and it has never run. A reverted `production_intermediate` configuration ($W = 5L$, $n_{\min} = 40$) was costed at ~ 2.2 h and also never run.

Proposed, never attempted

Extending the Classen-Howes Appendix B.6 recursion to an exact $\rho_F(\nu = 1/2)$ and an exact multiplicity table; the paired $W = 5L$ vs. $W = 10L$ disentangling test on identical trajectories (now largely superseded by the W -scan); density–density correlation functions, which are the README’s actual long-term target and have not been touched.

3 Results

3.1 H1: the frozen fraction is flat in N

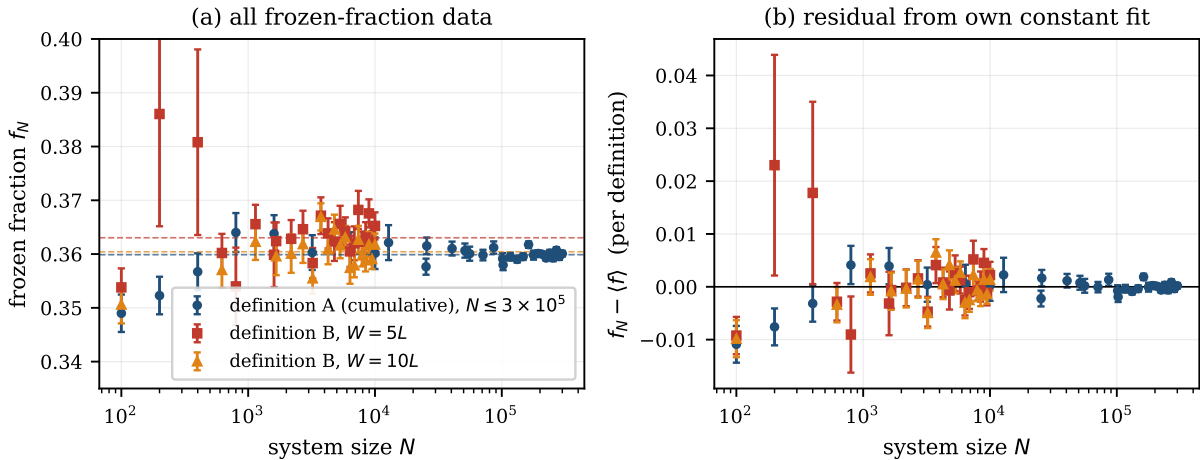


Figure 1: All frozen-fraction data in the project. (a) Definition A (archived, 31 points, $N = 100$ – 3×10^5), definition B at $W = 5L$ (24 points) and at $W = 10L$ (20 points); dashed lines are the inverse-variance weighted mean of each series. (b) The same data as residuals from each series’ own constant fit. The two high points with large error bars near $N = 200$ –400 come from the `test.yaml` smoke run, whose target relative error was 10% rather than 1%. Error bars are ± 1 standard error of the ensemble mean throughout.

Table 1 collects the model comparison, recomputed here from the raw `.npy` samples rather than copied from the log. The conventions are those already used in the project: weighted least squares with weights $1/\sigma^2$, $\chi^2 = \sum((f - \text{model})/\sigma)^2$.

Reading of Table 1.

- **H1c is excluded** under every definition, by three orders of magnitude in χ_ν^2 . This is not a marginal call.

Table 1: Model comparison for H1. “ α ” in H1b is the decay exponent, so $\alpha > 0$ means f_N vanishes. Negative fitted α means the best fit is a slight *increase* with N .

Dataset	Model	parameters	χ^2/dof	χ_ν^2
Definition A, 31 pts, $N = 100\text{--}3 \times 10^5$	H1a constant	$f_\infty = 0.35988 \pm 0.00015$	47.16/30	1.572
	H1a $+aN^{-\alpha}$	$f_\infty = 0.35990 \pm 0.00020, \alpha = 1.28 \pm 0.78$	33.36/28	1.192
	H1b $cN^{-\alpha}$	$\alpha = -0.00083 \pm 0.00050$	44.39/29	1.531
	H1c $c/\ln N$	$c = 4.218$	83289/30	2776
Definition B, $W = 5L$, 24 pts, $N = 100\text{--}10^4$	H1a constant	$f_\infty = 0.36303 \pm 0.00065$	24.71/23	1.074
	H1a $+aN^{-\alpha}$	$f_\infty = 0.36430 \pm 0.00240, \alpha = 0.63 \pm 0.63$	17.41/21	0.829
	H1b $cN^{-\alpha}$	$\alpha = -0.00443 \pm 0.00180$	18.62/22	0.846
	H1c $c/\ln N$	$c = 2.886$	9655/23	420
Definition B, $W = 10L$, 20 pts	H1a constant	$f_\infty = 0.36043 \pm 0.00057$	28.11/19	1.479
	H1c $c/\ln N$	$c = 2.935$	10318/19	543

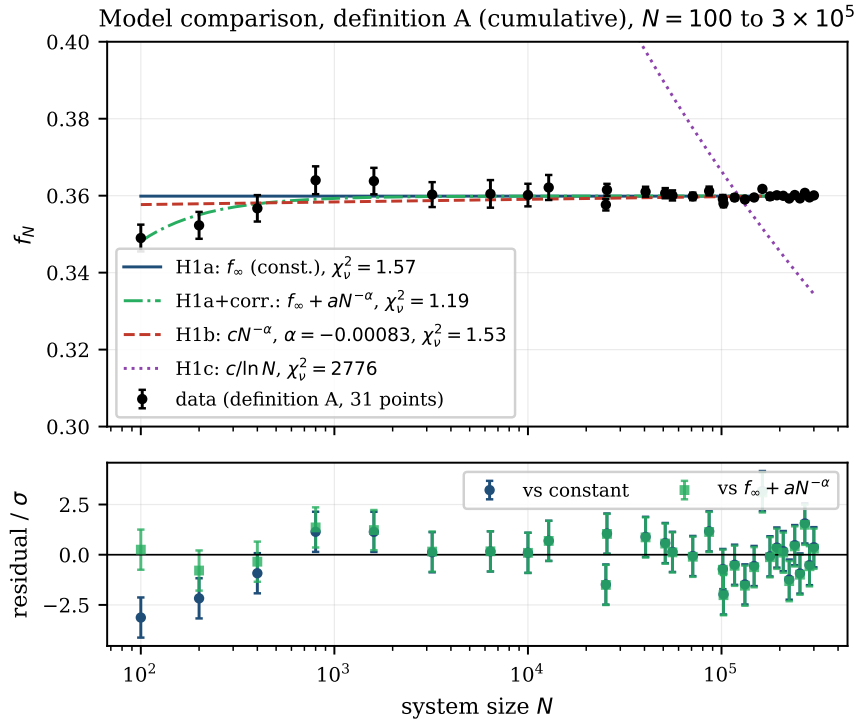


Figure 2: The four candidate forms fitted to the definition-A dataset (the widest range available, $N = 100$ to 3×10^5). H1c ($c/\ln N$, dotted) fails catastrophically. H1b’s best fit is indistinguishable from a constant because its fitted exponent is consistent with zero. Lower panel: residuals in units of σ .

- **H1b is not excluded as a functional form**, but its fitted exponent is consistent with zero and slightly negative under all three datasets; the 95% confidence interval from definition A is $\alpha \in [-0.0018, +0.00015]$. Taking the upper end at face value, f_N could fall by at most $\sim 0.03\%$ per decade of N . Over the four decades measured this is unobservable, and the fit is degenerate with a constant.
- **H1a fits.** For definition B at $W = 5L$ the constant model has $\chi^2_\nu = 1.07$ ($p = 0.37$): the data are entirely consistent with a constant. For definition A, $\chi^2_\nu = 1.57$ ($p = 0.024$) — a mild but real excess of scatter over the quoted error bars, discussed in Section 6.

3.2 The settings comparison, and why it is confounded

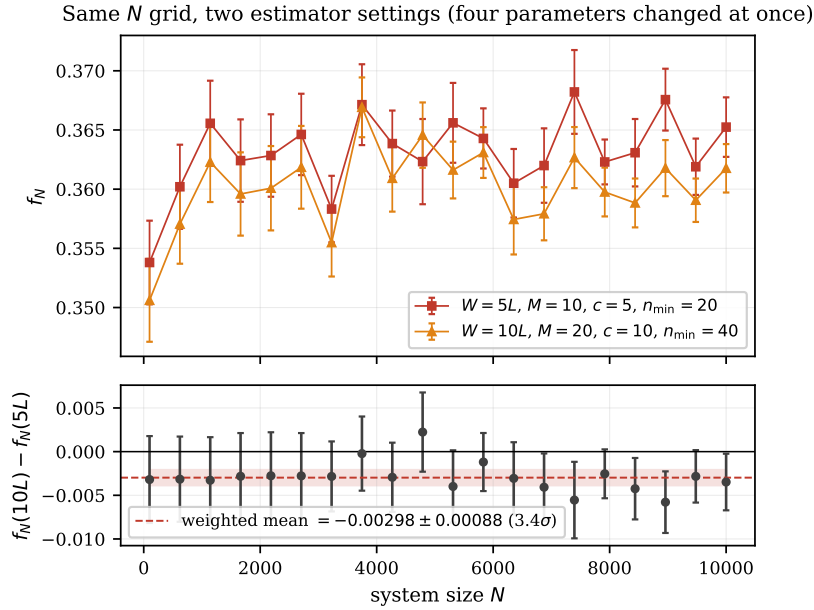


Figure 3: The 13 September re-run. Four parameters were changed simultaneously (W , M , $c_{\text{selfscale}}$, $n_{\min}$ all doubled). Lower panel: the shift is small, systematic and essentially N -independent.

The doubled-settings re-run shifted 19 of 20 points downward by a weighted mean of -0.00298 ± 0.00088 (3.4σ pooled; no individual point exceeds 1.96σ). The shift shows no dependence on system size (slope against $\ln N$: -0.00013 ± 0.00036 , $p = 0.73$).

Two of the four changed parameters independently predict a downward shift: doubling W must lower f_B by the monotonicity above, and doubling $c_{\text{selfscale}}$ lengthens burn-in. The other two (M , $n_{\min}$) only reduce variance. As designed, the experiment cannot separate these. The W -scan below settles it: the W effect alone accounts for 0.003–0.004, which is the entire observed shift, leaving nothing that requires a burn-in explanation.

3.3 W -dependence of definition B

Table 2: [new, this document] $f_B(W)$ from a single span of $200L$ steps per realisation. Standard errors are 0.0057–0.0084; because all columns come from the same trajectories, *differences* between columns are far more precise than these errors suggest.

W/L	1	2	5	10	20	50	100	200
$L = 250$	0.4067	0.3649	0.3505	0.3483	0.3474	0.3474	0.3474	0.3474
$L = 1000$	0.4187	0.3848	0.3662	0.3626	0.3625	0.3625	0.3625	0.3625
$L = 4000$	0.4184	0.3828	0.3686	0.3653	0.3648	0.3647	0.3647	0.3647

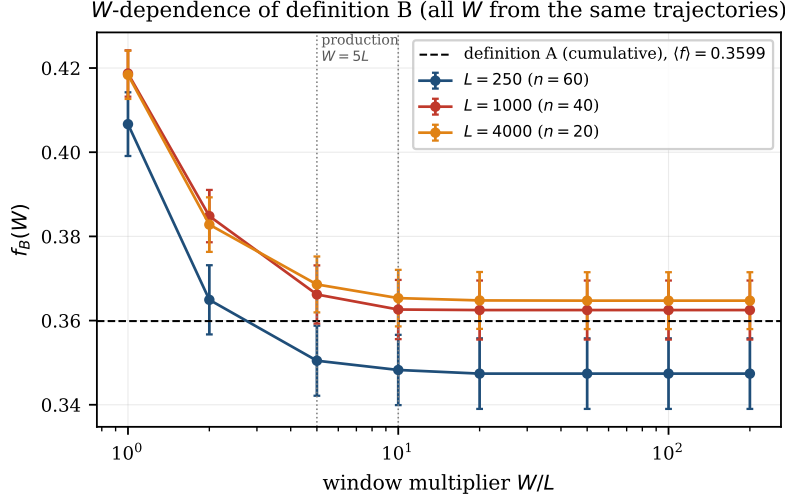


Figure 4: [new, this document] f_B as a function of the trailing-window length, with every W read off the *same* trajectories, so the W -dependence carries no realisation noise. The curve is flat to the fourth decimal from $W \approx 20L$ onward at all three L . Dashed line: the definition-A weighted mean over $N \geq 100$ for reference (a different estimator over a different N range, shown only to set the scale).

The production window $W = 5L$ overshoots the plateau by $+0.0031$, $+0.0037$ and $+0.0039$ at $L = 250, 1000, 4000$; $W = 10L$ by $+0.0009$, $+0.0001$, $+0.0006$. So the definitional freedom in H1 v2 costs at most about 1% of the value and does not grow with L .

I also measured the definition-A estimator’s effective observation window directly: $T_A = 26.7L$ at $L = 1000$ and $32.1L$ at $L = 4000$ (30 realisations each). Both lie inside the saturated regime of $f_B(W)$, which explains why definitions A and B give consistent answers at matched L despite being constructed differently.

3.4 Validation against exact enumeration

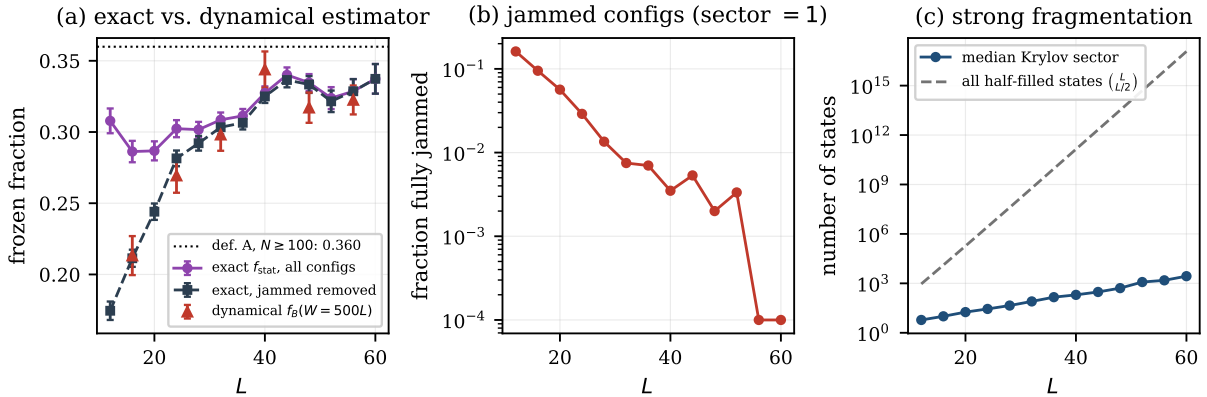


Figure 5: [new, this document] (a) The exact static frozen fraction from breadth-first enumeration of the Krylov sector, with and without fully-jammed configurations, against the dynamical estimator at saturated W . (b) The fraction of random half-filled configurations that are completely jammed (Krylov sector = one state), which is what forces the two curves in (a) apart at small L . (c) Median Krylov sector size against the total number of half-filled configurations.

Three things come out of this.

(i) **The estimator is unbiased where it can be checked.** Against ground truth, on the sub-ensemble it can sample, the dynamical estimator shows no detectable bias ($\chi^2_\nu = 0.83$). This

Table 3: [new, this document] Estimator validation. “Exact, jammed removed” is the sub-ensemble the dynamical estimator can actually sample, since burn-in reports jammed realisations as non-converged and discards them.

L	exact f_{stat} (all)	exact, jammed removed	dynamical $f_B(W=500L)$	difference
16	0.2863 ± 0.0075	0.2114 ± 0.0060	0.2132 ± 0.0137	$+0.0019 (+0.1\sigma)$
24	0.3023 ± 0.0060	0.2815 ± 0.0055	0.2694 ± 0.0120	$-0.0121 (-0.9\sigma)$
32	0.3086 ± 0.0051	0.3033 ± 0.0050	0.2981 ± 0.0113	$-0.0052 (-0.4\sigma)$
40	0.3274 ± 0.0047	0.3250 ± 0.0046	0.3438 ± 0.0127	$+0.0188 (+1.4\sigma)$
48	0.3347 ± 0.0060	0.3334 ± 0.0059	0.3171 ± 0.0106	$-0.0163 (-1.3\sigma)$
56	0.3286 ± 0.0085	0.3286 ± 0.0085	0.3225 ± 0.0102	$-0.0061 (-0.5\sigma)$
			$\chi^2 = 4.97$ on 6 points	$\chi^2_\nu = 0.83$

is the check H1 has been asking for since it was written.

(ii) **A real small- L artifact is identified.** Fully jammed configurations — where no window is active at all, so every site is trivially frozen — make up 16.2% of random half-filled configurations at $L = 12$, 9.5% at $L = 16$, 0.75% at $L = 32$ and 0% by $L = 56$. The dynamical estimator discards them, which biases it low by up to 4.7σ at $L = 16$. At production sizes ($L \geq 100$) every run reports `n_discarded`= 0, so the artifact is absent there — but note that this is an extrapolation of an exponentially decaying jamming probability, not a direct check at production L .

(iii) **f_{static} is still rising at $L = 60$.** It goes from 0.286 at $L = 16$ to 0.337 ± 0.010 at $L = 60$, so the exact small- L numbers cannot be compared directly with the 0.36 measured at $N \geq 100$; they are consistent with approaching it but do not by themselves establish the asymptote.

Strong fragmentation, measured. Fitting the median sector size gives $(1.135)^L$ against $\binom{L}{L/2} \sim (1.969)^L$, so a typical Krylov sector occupies a fraction $\sim (0.576)^L$ of its symmetry sector. This is a direct measurement of strong Hilbert-space fragmentation in *this* model, and it independently reproduces what the Classen-Howes thesis predicts for $(d, k) = (2, 4)$ at every filling.

3.5 H2: active-region length distribution

Table 4: H2 production run (`gap_length_gap_length_production.20260914T045549Z`), $W = 5L$, $c_{\text{selfscale}} = 5$, one post-burn-in snapshot per realisation, zero-length gaps excluded from $P_L(\ell)$. λ and α are full-range OLS fits of $\ln P$ against ℓ and against $\ln \ell$.

L	realisations	pooled gaps	$\langle \ell \rangle$	$P(\ell=0)$	λ	R_{exp}^2	α	R_{pow}^2
100	1400	5643	16.115	0.8850	12.95	0.952	2.42	0.879
200	720	5825	15.730	0.8888	13.26	0.917	2.49	0.872
400	320	5261	15.529	0.8864	13.27	0.939	2.54	0.897
800	160	5225	15.633	0.8872	13.22	0.943	2.43	0.895
1600	100	6544	15.554	0.8876	13.76	0.910	2.51	0.892
3200	60	7793	15.665	0.8886	13.08	0.966	2.60	0.904
6400	40	10416	15.636	0.8882	12.89	0.952	2.72	0.889
12800	20	10530	15.407	0.8877	14.15	0.915	2.72	0.901

What the numbers say. The exponential form is preferred over a power law at *every* L when both are fitted over the full range ($R_{\text{exp}}^2 = 0.910\text{--}0.966$ against $R_{\text{pow}}^2 = 0.872\text{--}0.906$). The decay length $\lambda \approx 12.9\text{--}14.2$ sites shows no systematic trend with L over two decades, as do $\langle \ell \rangle = 15.4\text{--}16.1$ and the zero-gap rate $0.885\text{--}0.889$.

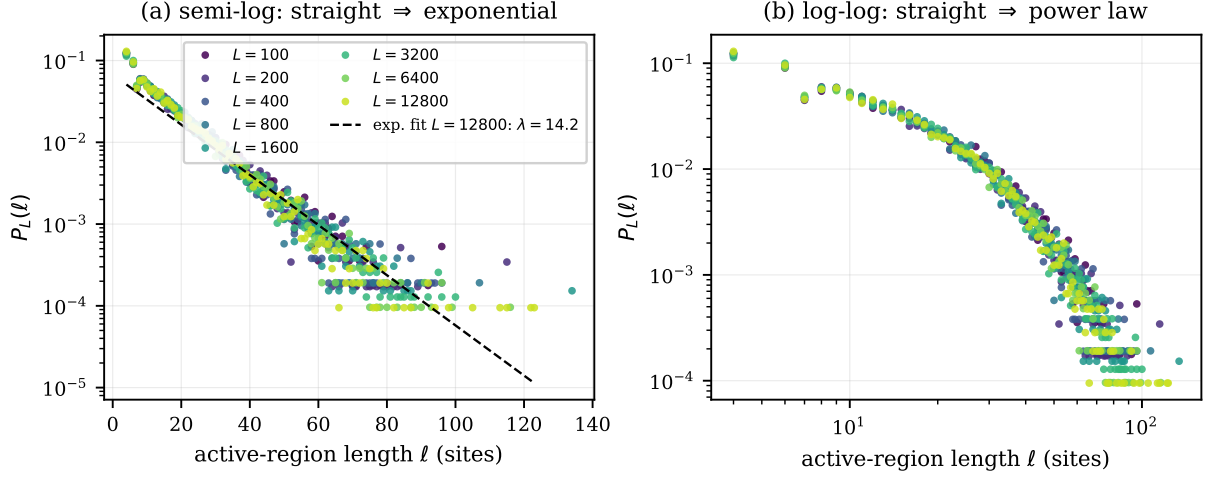


Figure 6: The measured length distribution on the two diagnostic axes. (a) Semi-log: the data are straight over more than two decades in P , which is the exponential signature. (b) The same data log-log, where a power law would be straight; it is visibly curved. The flat floor near $P \sim 10^{-4}$ in both panels is the single-count resolution limit ($1/n_{\text{pooled}}$), not physics.

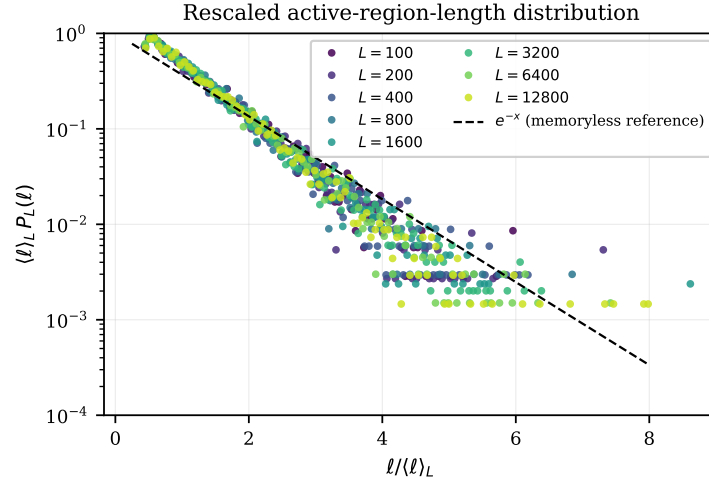


Figure 7: Rescaled distributions for all eight L . The collapse is the H2a signature: the shape is L -independent, and close to the memoryless reference e^{-x} .

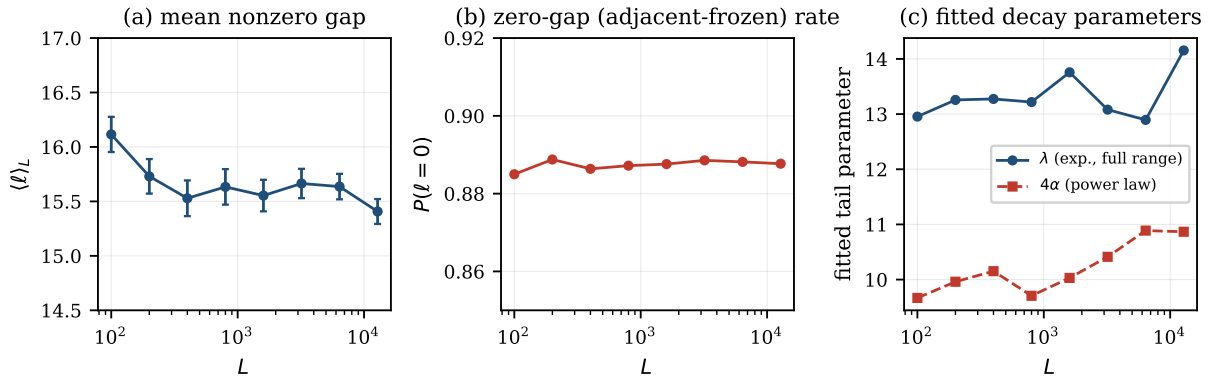


Figure 8: L -dependence of the summary statistics. All three are flat over two decades in L . In (c) the power-law exponent is scaled by 4 to share the axis.

A caveat the project already recorded: if the fit is restricted to the tail ($\ell \geq 15$) the two forms become nearly indistinguishable ($R_{\text{exp}}^2 = 0.876\text{--}0.954$ vs. $R_{\text{pow}}^2 = 0.897\text{--}0.951$, with the power law marginally ahead at five of eight L). The full-range comparison is the more discriminating one, because it includes the region around the peak where a power law would have to diverge and does not; but it is worth being explicit that the *far tail alone* does not by itself settle the question, and no likelihood-based model selection (AIC, or a likelihood-ratio test on the untruncated samples) has been done.

3.6 Consistency checks

Four independent checks were run while preparing this document; all pass.

1. **The dynamics conserves what it claims to.** [\[new, this document\]](#) Over 1.44×10^5 steps spanning $L = 40, 100, 400$ and 12 seeds each: magnetisation $\sum_i s_i$ was conserved exactly (0 failures), the dipole moment was conserved exactly modulo L (0 failures), and the incremental $O(1)$ active-site list agreed with a full brute-force rescan at every one of 720 spot checks. The unwrapped dipole was observed to take only the values $\{0, -L\}$ relative to its initial value, i.e. it is conserved exactly except for the expected jump of L when a move wraps the periodic boundary. This had never been tested in the repository.
2. **Cross-pipeline.** The H2 gap-length run records the snapshot frozen fraction as a by-product, computed by different code from the H1 estimator (single window rather than M averaged windows, separate seed stream). At the five overlapping L the two agree within -1.1σ to $+1.0\sigma$.
3. **Exact identity.** Since every frozen site opens exactly one gap on the ring, the mean over *all* gaps (zeros included) must equal $(1-f)/f$. Recomputing from the stored counts reproduces this to a ratio of 1.0000 at every L , confirming the gap extraction and the bookkeeping of discarded zeros are internally consistent.
4. **Estimator vs. ground truth.** Table 3.

3.7 The forbidden lengths: an exact argument

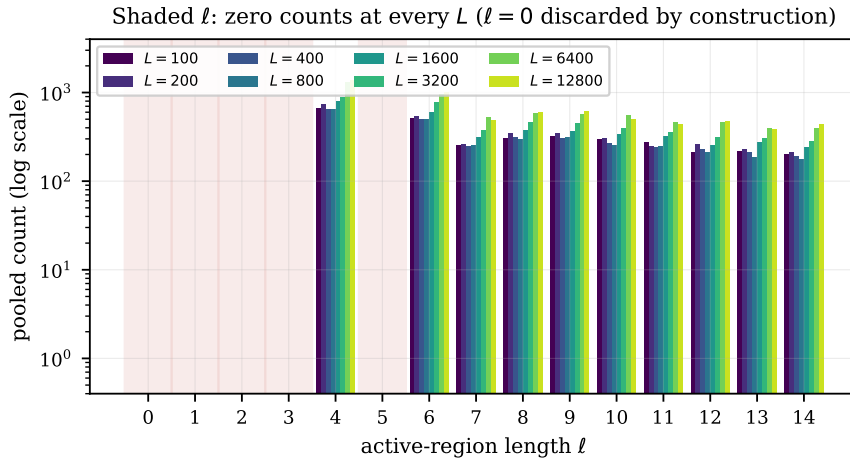


Figure 9: Pooled counts at small ℓ . Shaded columns have *exactly* zero counts at every L , across $\sim 57,000$ pooled gaps. $\ell = 0$ is excluded by construction; $\ell = 1, 2, 3, 5$ are absent dynamically.

Let $S \subseteq \mathbb{Z}_L$ be the set of window positions that fired at least once during the measurement window, and $F = \bigcup_{p \in S} \{p, p+1, p+2, p+3\}$ the set of sites recorded as active. A maximal run of F whose fired positions run from p_1 to p_k (consecutive gaps ≤ 4 , so the 4-blocks chain) covers $[p_1, p_k + 3]$, hence has length

$$\ell = p_k - p_1 + 4. \quad (3)$$

So $\ell \geq 4$ always, excluding $\ell \in \{1, 2, 3\}$, and $\ell = 5$ requires $p_k - p_1 = 1$: exactly the two positions p and $p + 1$ fired, and nothing else in that component.

Lemma 1 (verified exhaustively). *For every configuration of five consecutive sites: (i) window positions p and $p + 1$ are never simultaneously active; (ii) firing p leaves $p + 1$ inactive; (iii) firing $p + 1$ leaves p inactive.*

Verification. Direct enumeration of all $2^5 = 32$ configurations gives 0 violations of each of (i), (ii), (iii). For contrast, offsets 2, 3, 4 admit 2, 2, 4 simultaneously-active configurations respectively, which is what makes $\ell = 6, 7, 8$ reachable. $\square$

Proposition 1. *Under this dynamics, a maximal active run of length exactly 5 cannot occur, while $\ell = 4$ and every $\ell \geq 6$ can.*

Proof. Suppose a component of length 5 occurred; then only positions p and $p + 1$ fired in it, both at least once, and no other position within distance 4 fired (any such firing would lengthen the component). Note that window $p + 1$ spans sites $[p + 1, p + 4]$, all inside the component, and the only positions touching those sites are $p - 2, p - 1, p, p + 1$; the first two are excluded by the same argument, so only p and $p + 1$ can alter the activity of $p + 1$. Consider the first firing among $\{p, p + 1\}$, say p (the other case is symmetric using (iii)). At that instant p is active, so by (i) $p + 1$ is not. After the firing, by (ii), $p + 1$ is still inactive. Further firings of p only toggle its window between A and B , and (ii) applies to each; $p + 1$ therefore never becomes active and never fires, contradicting the assumption. Lengths 4 and ≥ 6 are realisable because a single position, or positions at offset 2, 3, 4, $\dots$, can fire. $\square$

This reproduces the observed support exactly: $\{4\} \cup \{6, 7, 8, \dots\}$. It is consistent with, and sharper than, the thesis result $\ell_{\min} = \ell_{\max} = 4$ for the minimal two-particle bubble, which explains $\ell \leq 3$ and the prominence of $\ell = 4$ but does not address $\ell = 5$.

Status of Proposition 1. The exhaustive verification of Lemma 1 is machine-checked and reproducible. The combinatorial argument built on it was written while preparing this document and *has not been reviewed by the researcher*. Per the project's own rule that theoretical arguments are confirmed before being relied on, it should be checked before being cited or built upon.

4 Claims, graded by evidence

Established — directly supported by data in this repository

- **[ESTABLISHED] Logarithmic decay (H1c) is excluded.** $\chi^2_\nu = 2776$ (definition A), 420 and 543 (definition B). *Evidence:* Table 1, Fig. 2.
- **[ESTABLISHED] Any power-law decay of f_N is bounded to $\alpha < 1.5 \times 10^{-4}$ at 95% confidence over $N = 10^2 - 3 \times 10^5$.** *Evidence:* H1b fit, definition A.
- **[ESTABLISHED] The frozen fraction is 0.359–0.363 depending on definition, with a definitional spread of ~ 0.004 .** *Evidence:* three independent weighted means; the ordering $f_A < f_B(10L) < f_B(5L)$ obeys the required monotonicity.
- **[ESTABLISHED] $f_B(W)$ saturates for $W \gtrsim 20L$;** the production $W = 5L$ overshoots by 0.003–0.004. *Evidence:* Table 2, Fig. 4.
- **[ESTABLISHED] The dynamical estimator reproduces exact static reachability** on the sub-ensemble it can sample, $\chi^2_\nu = 0.83$ over six L . *Evidence:* Table 3.
- **[ESTABLISHED] The model is strongly fragmented:** sector/symmetry-sector $\sim (0.576)^L$. *Evidence:* Fig. 5(c).
- **[ESTABLISHED] Active-region lengths 1, 2, 3, 5 cannot occur.** *Evidence:* zero counts in $\sim 57,000$ pooled gaps at eight L , plus the exhaustive Lemma 1 (the argument assembling them, Proposition 1, is pending review).
- **[ESTABLISHED] An exponential describes $P_L(\ell)$ better than a power law over the full range, at every L .** *Evidence:* Table 4, Fig. 6.

Strong observations — robust but needing more verification

- **[STRONG OBSERVATION]** f_N saturates to a nonzero constant (H1a). Flat over more than three decades, under two independent definitions, with an exclusion of the alternatives. What keeps this from “established” is that no measurement of a decay slower than $\alpha \approx 10^{-4}$ is possible with this range, and the underlying quantity is only *known* to be definition-independent up to the 0.004 spread.
- **[STRONG OBSERVATION]** The active-region length distribution is L -independent in shape (H2a over H2b). Collapse in Fig. 7; λ , $\langle \ell \rangle$ and $P(\ell=0)$ flat over two decades. Weakened by the fact that the far tail alone does not discriminate, and the largest L rests on only 20 realisations.
- **[STRONG OBSERVATION]** The settings shift is entirely a W effect. The W -scan accounts for 0.003–0.004 of the observed 0.00298; nothing is left for a burn-in explanation.

Conjectures / interpretations not established

- **[CONJECTURE]** $f_\infty \simeq 0.36$ is the thermodynamic-limit static frozen fraction. The exact calculation only reaches $L = 60$, where $f_{\text{static}} = 0.337 \pm 0.010$ and still rising; the identification of the measured plateau with the $L \rightarrow \infty$ static value is an extrapolation.
- **[CONJECTURE]** Because active regions are exponentially distributed and L -independent, global transport is arrested with no anomalous hydrodynamic tail. This is the README’s motivating inference, and the data support the premise — but the conclusion concerns density–density correlation functions, which have never been computed in this project.
- **[CONJECTURE]** The measured exponential is the thesis’s $\nu^x(1-\nu)^{\ell-x}$ law. Plausible and the forms match, but the thesis’s prediction has never been evaluated numerically at $\nu = 1/2$ and compared point-by-point against our $P_L(\ell)$.
- **[CONJECTURE]** $\ell = 5$ is forbidden at every particle number x . Proposition 1 proves it for the dynamical active runs we measure; the literature note records a separate hand-argument for $x = 3$ only, which was never verified against our data.

Claims with currently insufficient evidence

- Any statement about $N > 10^4$ under the current definition. The largest definition-B run is $N = 10^4$; everything beyond that is definition A.
- Any statement about densities other than $1/2$. Nothing else has been run, yet the literature’s whole framework is organised around filling.
- The precise value “ 0.36 ± 0.009 ” recorded in HYPOTHESES.md. The ± 0.009 has no stated derivation; the inverse-variance weighted errors are ± 0.0007 , and the definitional spread is ± 0.004 . The quoted uncertainty appears to be a conservative eyeball rather than a computed quantity.

5 The current scientific picture

Question. Does a dipole-conserving hardcore chain at half filling freeze a finite fraction of its sites in the thermodynamic limit, and are the surviving active regions microscopic or scale-free?

Approach. Direct stochastic simulation of the constrained dynamics, with the frozen set identified by a trailing-window criterion after a self-scaling burn-in; ensembles sized adaptively to a 1% relative error; and, at small L where it is possible, exact enumeration of the reachable configuration space as ground truth.

Results. f_N is flat at 0.359–0.363 from $N = 10^2$ to 3×10^5 . Logarithmic decay is excluded; power-law decay is bounded below 10^{-4} in exponent. The estimator is validated against exact enumeration. Active-region lengths are exponentially distributed with $\lambda \approx 13$ sites, L -independent, supported on $\{4\} \cup \{6, 7, \dots\}$.

Observation. Two independent structural facts coexist: a finite frozen backbone occupying $\sim 36\%$ of the chain, and active clusters whose size distribution has a fixed microscopic scale that does not grow with the system. Exact enumeration shows the accessible configuration space is an exponentially small fraction of the symmetry sector.

Interpretation (labelled as such). The chain is strongly Hilbert-space fragmented at half filling. Because the active regions have a finite correlation length, the system does not possess the rare large clusters that would be needed to carry long-range relaxation through a frozen background. In the README’s framing this is the “exponential” branch: transport is arrested, and no anomalous tail in the global density–density correlator is expected *from this mechanism*.

Claim. H1a (saturation) is supported and its competitors are excluded within the measured range; H2a (an L -independent, memoryless active-region distribution) is supported over H2b. The forbidden-length structure is a consequence of the 4-site move geometry rather than a statistical accident.

6 Contradictions, weaknesses and problems

This section is deliberately unflattering.

1. **HYPOTHESES.md H1 still defines the wrong quantity.** Its order-parameter paragraph defines f_N statically (“no sequence of allowed moves can ever change its occupation ... a property of the initial string alone, not of a stochastic trajectory or run time”), but every production number in the project is a dynamical proxy with an explicit run-time parameter. Section 3.4 now shows the two agree, but the document has never been corrected, and a reader taking H1 at face value would misunderstand what was measured. The **[UNCERTAIN]** block in H1 is also now out of date.
2. **The “ 0.36 ± 0.009 ” in HYPOTHESES.md is not a computed error bar.** Nothing in the repository derives ± 0.009 . The statistical error is ± 0.0007 and the definitional spread ± 0.004 . Carrying an unexplained uncertainty into a recorded claim is a hazard if it is later quoted.
3. **The settings comparison was a confounded experiment.** Four parameters were changed simultaneously; two of them independently predict the observed sign. It is resolved only retrospectively, by the W -scan. A one-at-a-time design was proposed at the time and not run.
4. **Heterogeneous precision is silently pooled.** The 24-point “definition B” dataset mixes `test.yaml` points (10% target relative error) with production points (1%). The two visible outliers at $N = 200, 400$ in Fig. 1 are exactly those. Inverse-variance weighting limits the damage, but the exploratory notebook’s headline “weighted mean over 24 points” is not a homogeneous dataset.
5. **Definition A’s constant fit is formally a poor fit.** $\chi^2_\nu = 1.57$ on 30 dof is $p = 0.024$. Either the error bars are underestimated — plausible, since each point’s error is a realisation-to-realisation standard error that ignores any systematic residual burn-in bias — or there is real residual structure. The project investigated the latter (commensuration, second correction term, log-periodicity) and found nothing statistically supported, which leaves mildly underestimated uncertainties as the most likely explanation. That possibility has never been tested directly.

6. **Sampling at the largest sizes is thin.** At $L = 12800$, H2 rests on 20 realisations; the H1 $W = 5L$ run has $n = 20$ at eleven of twenty N -values. Adaptive sampling stops at a target *relative* error of the mean, which at large L is met almost immediately because realisation-to-realisation variance shrinks — so the largest, most physically interesting sizes get the *fewest* samples. This is a structural feature of the stopping rule worth reconsidering.
7. **A real discard bias exists and is only extrapolated away at production sizes.** Jammed configurations are discarded. This is a 4.7σ effect at $L = 16$. It is believed negligible at $L \geq 100$ because `n_discarded` = 0 in every production run and the jamming probability decays exponentially — but “zero discards observed” with $n \leq 1400$ only bounds the jammed fraction at $\lesssim 10^{-3}$, it does not measure it.
8. **The $W \rightarrow \infty$ plateau is a plateau over one decade only.** Flatness of $f_B(W)$ for $W \in [20L, 200L]$ does not prove convergence to the static value; a site with a flip timescale $\gg 200L$ would be invisible. The small- L agreement with exact enumeration is reassuring, but at $L \leq 56$, where absolute timescales are tiny.
9. **The tail-only fit does not support the headline H2 claim, and this asymmetry is under-emphasised.** Restricted to $\ell \geq 15$, the power law is marginally *better* at five of eight L . The full-range comparison is more discriminating for good reasons, but the honest statement is “exponential over the full range; the far tail alone is inconclusive”, not “exponential, confirmed”. No AIC or likelihood-ratio test has been performed on the raw samples.
10. **The archived analysis notebooks are dead code.** The three `h1_fit_frozen_fraction*.ipynb` notebooks are pinned to paths that have moved to `data/archive/`; they cannot be re-executed. The definition-A fits quoted throughout this document were therefore recomputed from the raw `.npz` files rather than trusted from notebook output. (They reproduce the logged values.)
11. **Stale statements in the research log.** The entry of 13 Sep states that `production_large_scale.yaml` is “scheduled to launch automatically at 2026-09-14 00:00”. It never ran, and the config has since been retargeted twice. Log entries are a historical record, so this is not wrong in itself, but nothing in the log records the interruption.
12. **The core dynamics had never been tested, and there is still no regression test.** Every number in the project flows through one `DipoleChain` class, whose conservation laws were asserted only in its docstring. I ran that check ad hoc while preparing this document and it passes cleanly (Section 3, check 1), and the independently written BFS code agrees on the move rule — so this is no longer an open risk. But the check exists only in this document’s history: there is no test file in the repository, so nothing will catch a future regression.
13. **The project’s stated physical target has not been approached.** The README’s actual goal is the density–density correlation function and its possible anomalous tail. Two structural inputs have now been measured, but no correlation function, no transport coefficient and no dynamical exponent has been computed.

7 Highest-value next steps, in priority order

1. **Commit the conservation-law check as a regression test.** (Minutes.) The check described in Section 3 passes, but it lives nowhere in the repository. Turning it into a test file — assert $\sum_i s_i$ and $\sum_i i n_i \bmod L$ invariant under `step()`, and the incremental active-list update equal to a full rescan — costs almost nothing and protects every number the project produces. Highest value-per-cost in the list.
2. **Run `production_large_scale.yaml`.** (4–5 h, already costed and approved.) It is the only planned measurement that extends definition B beyond $N = 10^4$, and the H1 claim currently has a one-decade gap between where the current definition stops (10^4) and where

the superseded one reached (3×10^5) .

3. **Re-weight the sampling toward large L .** Replace, or supplement, the relative-SE stopping rule with a minimum sample count that *grows* with L . At present the largest systems are the least sampled, which is backwards for a finite-size study.
4. **Do proper model selection on H2 with likelihoods.** Fit the raw ℓ samples by maximum likelihood to a geometric/exponential and to a discrete power law (with the $\ell \in \{4\} \cup \{6, \dots\}$ support handled explicitly), and compare with AIC and a likelihood-ratio test. The current R^2 -on-binned-counts comparison is weaker than the data deserve and is sensitive to the binning and to the single-count floor.
5. **Evaluate the thesis’s exact prediction at $\nu = 1/2$ and overlay it on Fig. 6.** The formula $\propto \nu^x(1 - \nu)^{\ell-x}$ plus $P_b(\nu)$ is closed-form; computing it and plotting it against our measured $P_L(\ell)$ converts a qualitative “consistent with the literature” into a quantitative test with no new simulation.
6. **Push the exact enumeration further with symmetry reduction.** $L = 60$ took 6.7 minutes; using translation symmetry and canonical forms should reach $L \approx 80$ –100 and would let $f_{\text{static}}(L)$ be extrapolated against the dynamical plateau directly rather than qualitatively.
7. **Measure the density–density correlation function.** This is the README’s actual target and the only way to convert the structural results into the physical claim the project wants to make.
8. **Vary the density.** Everything is at $\nu = 1/2$. The literature frames this model entirely in terms of filling, and the thesis’s claim that $(d, k) = (2, 4)$ is strongly fragmented *at every* filling is directly testable with the existing pipeline by changing one config value.

8 Questions I could not resolve from the project

1. **Which frozen-fraction number is the one you want to quote?** There are now three defensible candidates: 0.35988 ± 0.00015 (definition A, widest N range), 0.36303 ± 0.00065 (definition B at the production $W = 5L$), and the W -saturated plateau ≈ 0.3647 at $L = 4000$. They differ by more than their statistical errors. I have not picked one as authoritative anywhere in this document.
2. **Where does the ± 0.009 in HYPOTHESES.md come from?** I could not reconstruct it from any file. If it was meant as a conservative band covering the definitional spread, it should say so; if it was meant as an error bar, it disagrees with the computed one by an order of magnitude.
3. **Should “frozen” ultimately mean static or rolling?** They now agree numerically, so this is a presentational choice — but it changes what H1 claims. The static definition connects directly to the Hilbert-space-fragmentation literature; the rolling one is what is actually measurable at large L .
4. **Is discarding zero-length gaps the right convention for the physics you care about?** It removes $\sim 89\%$ of all gaps and breaks the $\langle \ell \rangle = (1 - f)/f$ identity. For a correlation-function calculation the full distribution including zeros is what enters. Was the intent to characterise the shape of nonzero clusters only?
5. **Is the README’s “aperiodic” a typo for “a periodic”?** The code is unambiguously periodic, and everything in the project assumes PBC, so I have read it as a typo — but if an aperiodic or quasiperiodic variant was ever intended, none of the current work addresses it.
6. **Do you want Proposition 1 checked independently before it is used?** Per your own rule I have flagged it rather than treating it as established.

Your suggestions / things you should consider

These are suggestions, not results.

- **Measure the distribution of frozen-run lengths too, not just active-run lengths.** You currently characterise the gaps but not the backbone. The frozen runs are the complementary structure, they are trivially available from the same snapshot (the code already builds the mask), and their distribution is what determines whether the frozen backbone is percolating-like or fragmented. Given $P(\ell = 0) \approx 0.888$ the frozen runs are clearly long, and nobody has looked at them.
- **The zero-gap rate is suspiciously constant.** 0.8850 to 0.8888 across two decades of L , i.e. variation in the fourth decimal. Combined with $\langle \ell \rangle$ and λ also being flat, this smells like an exactly computable constant of the model rather than an empirical number. It may be derivable from the same window-geometry argument as Proposition 1.
- **Exploit the “all W from one trajectory” trick more widely.** It converted a noisy 4-parameter comparison into a clean paired measurement at a fraction of the cost. The same pattern — record per-site last-flip times once, derive many observables — would give you the autocorrelation function $C(t)$ and the distribution of single-site flip waiting times essentially for free from runs you are already doing. The waiting-time distribution in particular would tell you whether the W -plateau is genuine or hiding a slow tail.
- **Consider reporting $1 - f$ rather than f .** The active fraction 0.64 is the quantity that enters transport arguments, and error propagation is more transparent there.
- **A theoretical connection worth checking.** Your active regions are bounded by frozen sites and have a minimum size of 4 with 5 forbidden. That is the signature of a constrained-tiling problem. The Moudgalya–Motrunich technique in your notes (canonical form $\rightarrow$ tiling $\rightarrow$ transfer matrix) would plausibly yield $P(\ell)$ in closed form for this model, including the exact λ , and you already have the numerical answer to check it against. This looks like the single most tractable piece of analytics available to you right now.
- **Be careful before abandoning the model.** The README says you would move on if the distribution is exponential. It is — but the exponential decay length $\lambda \approx 13$ sites is large compared to the 4-site move, and 89% of frozen sites are adjacent to another. Before concluding the model is uninteresting for hydrodynamics, it is worth checking whether the *frozen* structure has long-range correlations even though the active clusters do not.
- **Housekeeping.** The archived notebooks cannot be re-executed; either repoint them at `data/archive/` or add a line to their README stating they are frozen artifacts. And the two configured-but-never-run jobs should either be run or removed from the configs directory, since a config that has never executed is a trap for a future reader.

Figures regenerable via `python numerics/scripts/make_progress_figures.py`. Numerical values dumped to `numerics/data/processed/progress_numbers.json`. New measurements made for this document live under `numerics/data/raw/` in the `static_frozen_*` and `w_dependence_*` run directories, and are logged in `RESEARCH_LOG.md` (14 September entries). Document source: `progress.tex`.